

Specifiche algebriche per la stringa:

Specifiche sintattiche:

sorts: string, char, integer, boolean

operations:

new() $\rightarrow$ string

append(string, string) $\rightarrow$ string

add(string, char) $\rightarrow$ string

length(string) $\rightarrow$ integer

isEmpty(string) $\rightarrow$ boolean

equal(string, string) $\rightarrow$ boolean

crea nuove stringhe

concatena due stringhe

aggiunge un carattere a fine stringa

calcola la lunghezza di una stringa

predica se la stringa è vuota

predica se due stringhe sono uguali

I costruttori sono new() e add(string, char)

Specifiche semantiche:

dec ene $str: \text{string}, c: \text{char}, n: \text{integer}$

- $\text{length}(\text{new}()) = 0$
- $\text{length}(\text{add}(str, c)) = \text{length}(str) + 1$
- $\text{isEmpty}(\text{new}()) = \text{true}$
- $\text{isEmpty}(\text{add}(str, c)) = \text{false}$
- $\text{append}(\text{new}(), \text{new}()) = \text{new}()$
- $\text{append}(\text{new}(), \text{add}(str', c')) = \text{add}(str', c')$
- $\text{append}(\text{add}(str, c), \text{new}()) = \text{add}(str, c)$
- $\text{append}(\text{add}(str, c), \text{add}(str', c')) =$
 $= \text{add}(\text{append}(\text{add}(str, c), str'), c')$
- $\text{equal}(\text{new}(), \text{new}()) = \text{true}$
- $\text{equal}(\text{new}(), \text{add}(str', c')) = \text{false}$
- $\text{equal}(\text{add}(str, c), \text{new}()) = \text{false}$
- $\text{equal}(\text{add}(str, c), \text{add}(str', c')) = c = c' \text{ and } \text{equal}(str, str')$